

Siddhant Aggarwal

Heidelberg, Germany — +49 15510282126

siddhantaggarwal2001@gmail.com

LinkedIn — GitHub — Portfolio

Date of Birth: 25 Dec 2001 — **Nationality:** Indian



Profile

Motivated Master's student in Information Technology (AI Specialization) at SRH Heidelberg. Solid grounding in Computer Engineering, hands-on skill in Machine Learning, Battery Analytics, and Data Engineering. Experienced with technical documentation and international research teams. Seeking working student, internship, or entry-level AI/Data Science role for industrial R&D projects in Germany.

Profil

[Motivierter Master-Student im Bereich Informationstechnologie (AI-Spezialisierung) an der SRH Heidelberg. Praktische Erfahrung in Machine Learning, Batteriediagnostik und Datenanalyse mit Schwerpunkt auf industriellen und Forschungsanwendungen. Teamarbeit, internationale Projektumgebung und technische Dokumentation sind meine Stärken. Ich suche eine Werkstudenten- oder Praktikumsstelle im Bereich KI/ML oder Data Science, um mein Fachwissen in deutschen Industrie- und Forschungsprojekten einzubringen.]

Education

M.Eng. (Information Technology, AI Specialization)

SRH Hochschule Heidelberg, Germany

Expected Mar 2027

GPA: 2/5

B.E. Computer Engineering

Thapar Institute of Engineering and Technology, India

2024

Technical Skills

- **Programming:** Python, R, C++, SQL, MATLAB
- **Frameworks/Libraries:** PyTorch, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib
- **Tools:** Git, Streamlit, Jupyter, VS Code, Omnet++, Linux
- **Concepts:** Data Cleaning, Data Analysis, Data Visualization, Feature Engineering, Machine Learning, Battery Analytics

Experience

Data Analytics Intern

Centre for Railway Information Systems (CRIS), India

Jan 2024 – Jun 2024

- Cleaned and preprocessed large-scale datasets using Python and SQL, improving data quality for analysis.
- Developed interactive dashboards to visualize operational bottlenecks and assist strategic decision-making.
- Automated routine data auditing and reporting workflows, reducing manual workload and increasing team efficiency.
- Collaborated with cross-disciplinary teams to communicate data-driven insights clearly.

Key Projects

Battery Remaining Useful Life (RUL) Prediction

Python, PyTorch, Scikit-learn, EMD

- Built an end-to-end ML workflow for battery RUL prediction using NASA/CALCE data.
- Extracted and cleaned raw battery sensor data for predictive modeling
- Engineered features using statistical methods and Empirical Mode Decomposition (EMD) to capture signal nuances
- Optimized models with GridSearchCV and Optuna.
- Developed Streamlit dashboard for real-time prediction and uncertainty quantification.
- Link to the Dashboard web app: batteryhealth.streamlit.app
- Published findings and code: github.com/Siddhant2512/Intelligent-Battery-Monitoring

Handwritten Digit Classification (Python, Scikit-learn)

- Developed a Convolutional Neural Network model, Random Forest Classifier, n-gram model as well as Naive Bayes classifier for MNIST dataset; led model evaluation.

Communication Networks Simulation (Omnet++)

- Simulated Go-Back-N protocol and Token Bucket Traffic Shaper to assess reliability.

Image Processing (Python, OpenCV)

- Applied adaptive thresholding and HOG object detection.

TOPSIS Decision-Making (Python)

- Created and published multi-criteria decision package for ranking solutions.

Certifications

- Python for Data Science, AI & Development – IBM (Coursera)
- SQL for Data Science – UC Davis (Coursera)
- Tools for Data Science – IBM (Coursera)

Languages

English (C2), German (A2, progressing)